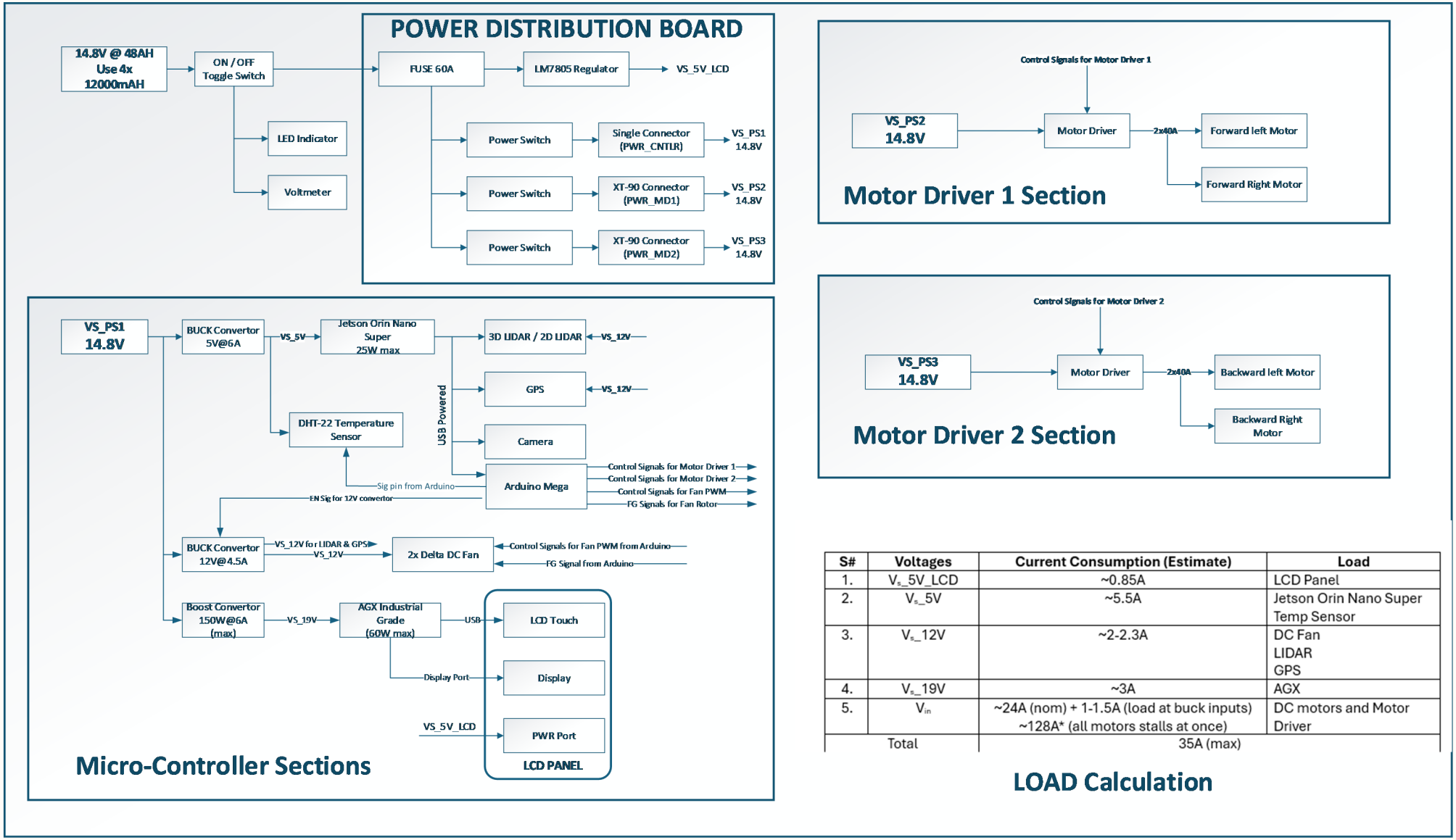
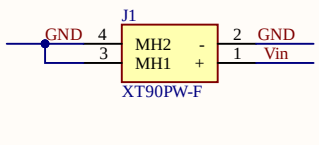


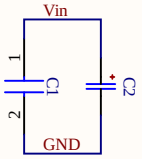
S#	Voltages	Current Consumption (Estimate)	Load
1.	V_s_5V_LCD	~0.85A	LCD Panel
2.	V_s_5V	~5.5A	Jetson Orin Nano Super Temp Sensor
3.	V_s_12V	~2-2.3A	DC Fan LIDAR GPS
4.	V_s_19V	~3A	AGX
5.	V_{in}	~24A (nom) + 1-1.5A (load at buck inputs) ~128A* (all motors stalls at once)	DC motors and Motor Driver
Total		35A (max)	

LOAD Calculation

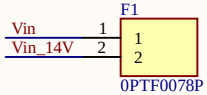
Title	*Block Diagram of System*	Designed by:	
Size: A4	Number:1	Revision:1.0	*M. Huzaifa Rahim Malik*
Date: 28/07/2025	Time: 12:07:34 pm	Sheet 1 of 2	*AM-Tech*
File: D:\NCP\Robotic Cars\Outdoor Vehicle\Protection Block\Schematic\Outdoor_Vehicle\Block.SchDoc			AITEC NCP ISB

Input Power Source

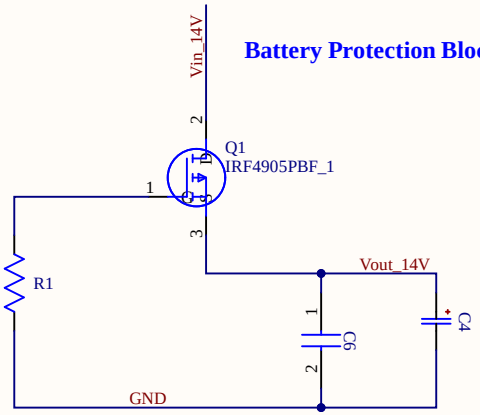




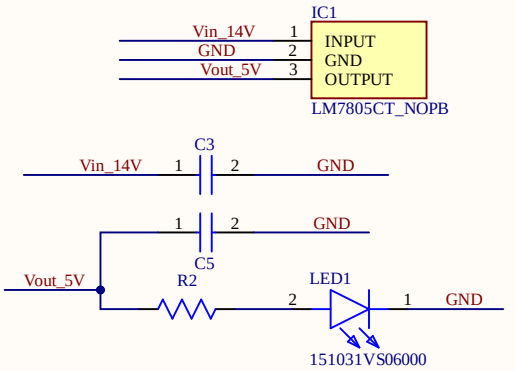
Fuse

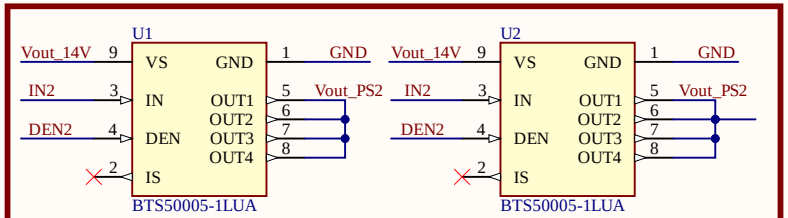


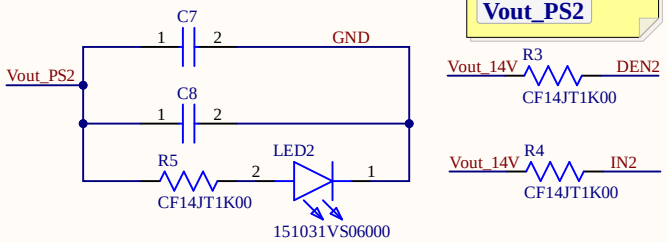
Battery Protection Block

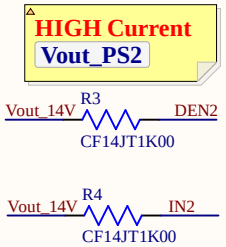


Voltage Regulator

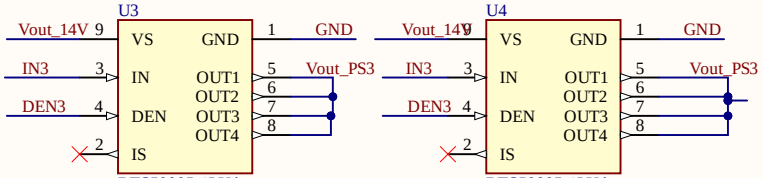


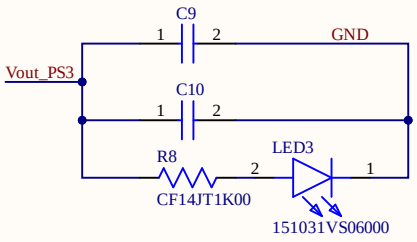


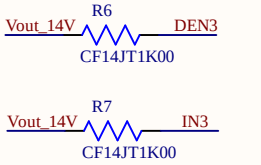




Power Switch for Motor Driver 1

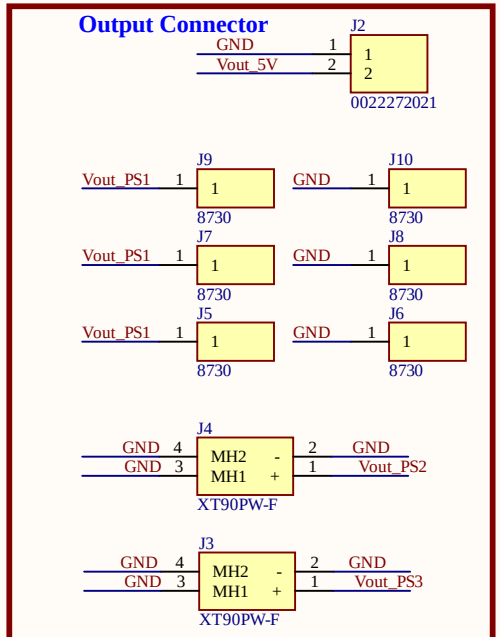


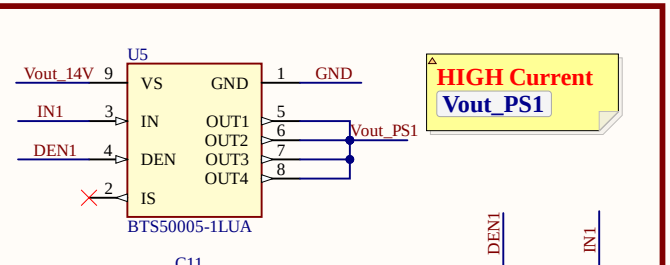


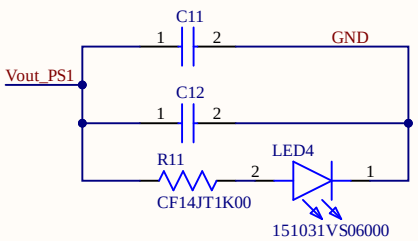


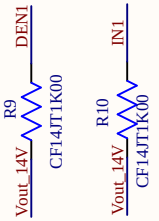
Power Switch for Motor Driver 2

Output Connector



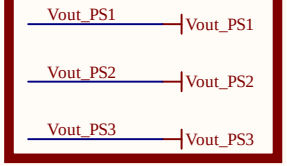




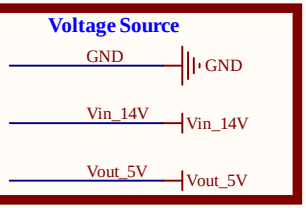


Power Switch for Micro Controller Supply

Output Voltages



Voltage Source



Title *Power Distribution Board*			**Designed by:** *M. Huzaifa Rahim Malik* *AM-Tech*
Size: A4	Number:2	Revision:1.0	*AITEC NCP ISB*
Date: 28/07/2025	Time: 12:07:35 pm	Sheet 2 of 2	
File: D:\NCP\Robotic Cars\Outdoor Vehicle\Protection Block\Schematic\Outdoor_Vehicle\Power Distribution Block.SchDoc			

Description	Designator	Manufacturer Name	Manufacturer Part Number	Supplier	Supplier Part Number	Value	Tolerance	Rating	Quantity
0.1 µF ±20% 50V Ceramic Capacitor Z5U Radial	C1, C5, C6, C7, C8, C9, C10, C11, C12	KEMET	C322C104M5U5TA	Digikey	399-4331-ND	0.1 µF	±20%	50V	9
CAP ALUM 470UF 20% 50V RADIAL TH	C2, C4	Nichicon	UHE1H471MHD3	Digikey	493-1615-ND	470 µF	±20%	50 V	2
CAP CER 0.33UF 50V X7R RADIAL	C3	TDK Corporation	FA24X7R1H334KNU06	Digikey	445-180638-1-ND	0.33 µF	±10%	50V	1
FUSE BLOCK CARTRIDGE PCB	F1	LITTELFUSE	0PTF0078P	Digikey	F6094-ND				1
IC REG LINEAR 5V 1.5A TO220-3	IC1	Texas Instruments	LM7805CT/NOPB	Digikey	296-47192-ND	5-35V		1.5A	1
Connector: DC supply; socket; male; XT90; 40A; 500V; PIN: 2; straight	J1	Amass	XT90PW-M	TEM Electronics Component	XT90PW-M	Male			1
CONN HEADER VERT 2POS 2.54MM	J2	Molex	0022272021	Digikey	WM4111-ND				1
Connector: DC supply; socket; female; XT90; 40A; 500V; PIN: 2; THT	J3, J4	Amass	XT90PW-F	TEM Electronics Component	XT90PW-F	Female			2
TERM BINDING POST M2.5 1 PIN PCB	J5, J6, J7, J8, J9, J10	Keystone Electronics	8730	36-8730-ND	Digikey				6
LED GREEN DIFFUSED 3MM ROUND T/H	LED1, LED2, LED3, LED4	Wurth Elektronik	151031VS06000	Digikey	732-5008-ND	2.2V		20mA	4
MOSFET P-CH 55V 74A TO220AB	Q1	Infineon	IRF4905PBF	Digikey	IRF4905PBF-ND			74A	1
RES 10K OHM 5% 1/4W AXIAL	R1, R2	Stackpole Electronics Inc	CF14JT10K0	Digikey	CF14JT10K0TR-ND	10 kOhms	±5%	0.25W, 1/4 W	2
RES 1K OHM 5% 1/4W AXIAL	R3, R4, R5, R6, R7, R8, R9, R10, R11	Stackpole Electronics Inc	CF14JT1K00	Digikey	CF14JT1K00TR-ND	1 kOhms	±5%	0.25W, 1/4W	9
Smart High-Side Power Switch	U1, U2, U3, U4, U5	Infineon Technologies	BTS500051LUAAUMA1	Digikey	448-BTS500051LUAAUMA1TR-ND	3.1V ~ 28V		65A	5

Board Stack Report

Stack Up		Layer Stack			
Layer	Board Layer Stack	Name	Material	Thickness	Constant
1		Top Overlay		0mil	
2		Top Solder	Solder Resist	0.4mil	3.5
3		Top Layer		2.756mil	
4		Dielectric 2	PP-006	11.811mil	4.1
5		GND	CF-004	2.756mil	
6		Dielectric 1	FR-4	27.559mil	4.8
7		PWR	CF-004	2.756mil	
8		Dielectric 3	PP-006	11.811mil	4.1
9		Bottom Layer		2.756mil	
10		Bottom Solder	Solder Resist	0.4mil	3.5
11		Bottom Overlay		0mil	
	Height : 63.005mil				

120.00mm

84.49mm

95.00mm

130.00mm

PWR_LCD

J2

C2

C2

F1

C6

Batt_In

R3

R4

Q1

C4

R2

R2

C5

C3

IC1

LED1

R7

R1

R1

R7

R6

R6

C1

U3

C10

U1

C8

C9

U4

U2

C7

U5

J3 PWR_MD2

R8

R8

LED3

R10

R9

R10

R9

R11

R11

C12

PWR_MD1

J4

R5

R5

LED2

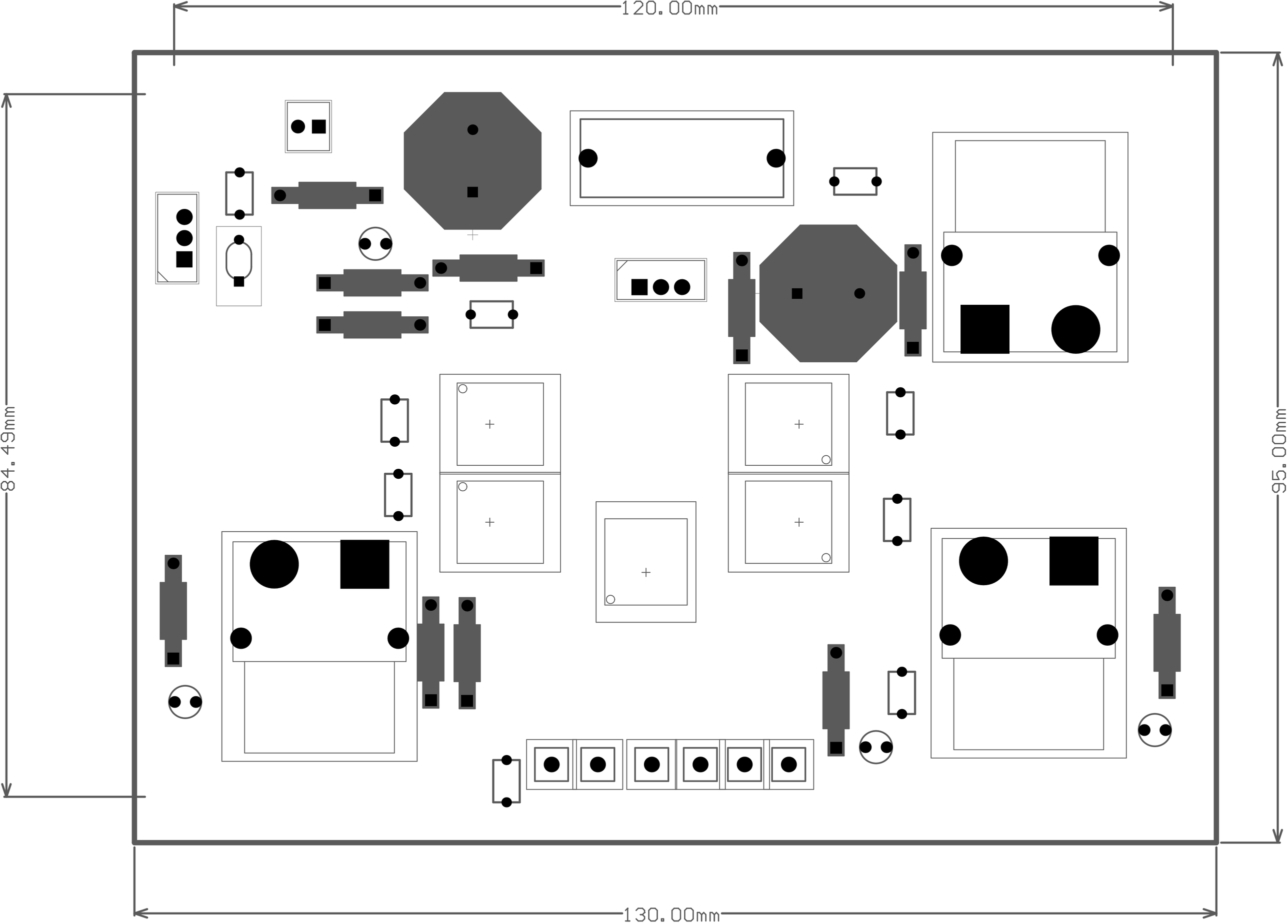
LED4

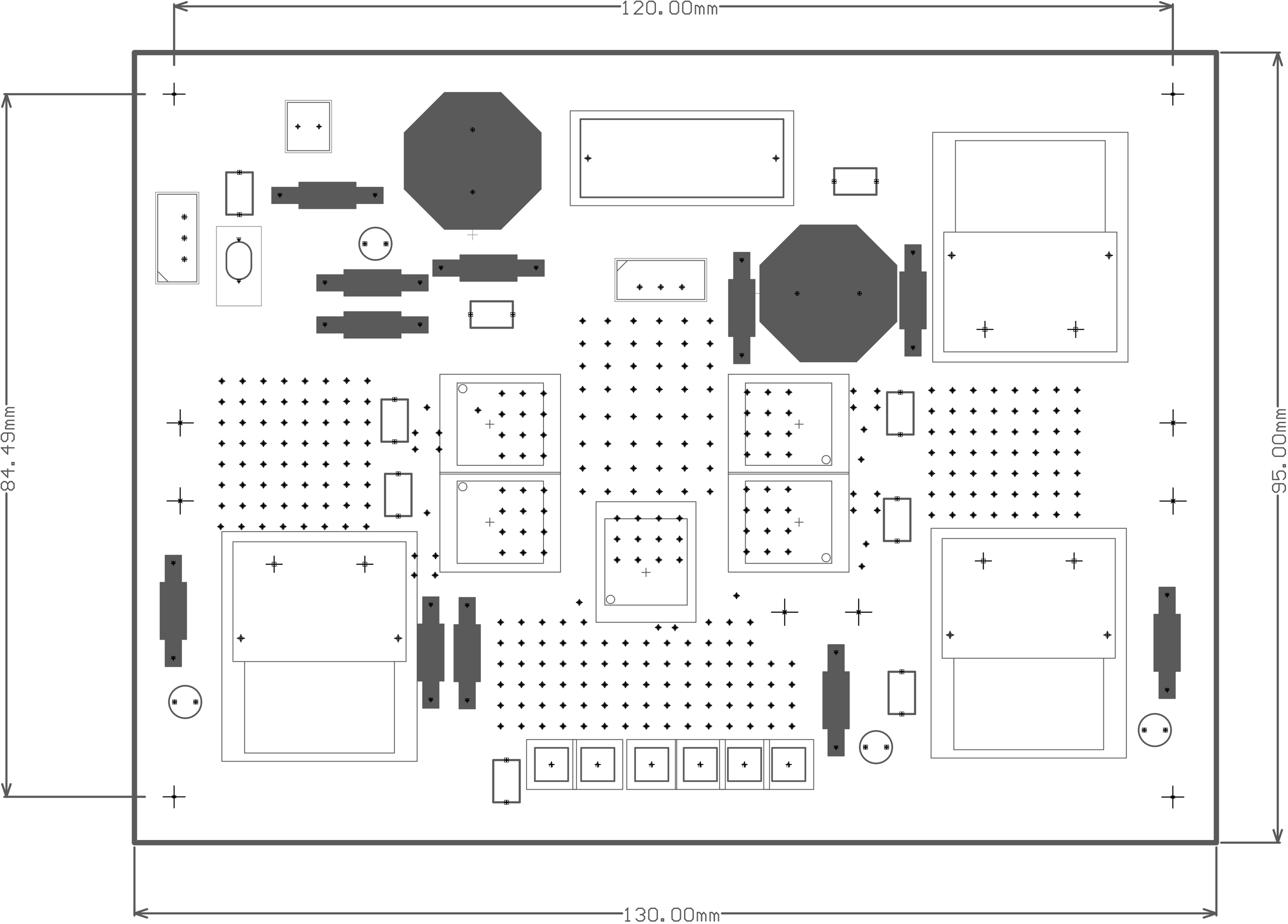
J5 J6 J7 J8 J9 J10

PWR_CNTRLER

C11

ATTeC, NCP Outdoor
Vehicle Car V1.0
Power Distribution
Board


120.00mm
84.49mm
95.00mm
130.00mm


120.00mm
84.49mm
95.00mm
130.00mm